

1. Administrative & Study Identification

Full Study Title: A Three-year, Multi-center, Double-blind, Extension Study to Evaluate the Long-term Safety and Efficacy of Ligelizumab in Patients Who Completed Ligelizumab's Phase III Studies in Food Allergy **Short Title/Acronym:** Long-term Extension Study of Ligelizumab in Food Allergy **Protocol Number:** CQGE031G12303B **NCT Number:** NCT05678959 **Sponsor Name:** Novartis **Investigational Product:** Ligelizumab **Phase of Development:** Phase III **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To evaluate the long-term safety and efficacy of ligelizumab (120 mg and 240 mg) in patients who have completed ligelizumab's Phase III core studies in food allergy. **Secondary Objectives:** To assess the long-term impact on Health-Related Quality of Life (HRQoL) using the Food Allergy Quality of Life Questionnaire - Teenager Form (FAQLQ-TF) and the Food Allergy Independent Measure (FAIM-TF/AF).

2.2 Background & Rationale Ligelizumab is a humanized IgG1 monoclonal antibody with a high affinity for IgE, designed to prevent IgE from interacting with high-affinity FcεRI receptors on mast cells and basophils. This extension study aimed to evaluate whether sustained treatment over up to three years would safely maintain or increase the threshold of allergic reactivity, ultimately protecting patients from life-threatening anaphylactic reactions due to accidental allergen exposure. *(Note: This trial was terminated early based on efficacy data reviews from the core study).*

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Interventional **Control Type:** Placebo-controlled (matching placebo used to protect dose blinding) **Blinding:** Double-blind **Randomization:** Rollover from a randomized Phase III core study

3.2 Planned Enrollment & Duration **Target \$N\$:** 163 participants (Actual Enrollment) **Number of Sites:** Approximately 50 locations **Estimated Study Start:** 27-Apr-2023 **Estimated Primary Completion:** 06-Mar-2025 (Terminated)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Signed informed consent and/or assent obtained prior to enrolling in the rollover study. 2. Participants had completed the treatment period in any of ligelizumab's Phase III studies in food allergy. 3. Willingness to adhere to the study visits and procedures, including receiving injections and participating in an open-label oral food challenge (OFC). 4. Agreement to continue avoiding exposure to core study allergens and any other foods they were allergic to.

4.2 Exclusion Criteria 1. Development of a severe or life-threatening episode of an allergic reaction that required intubation and/or intensive care unit (ICU) admission during the core studies. 2. Development of a serious adverse event which was suspected to be related to the study treatment judged by the investigator during the core study. 3. Development of uncontrolled asthma during the core study that compromised the safety of the participants.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Ligelizumab 120 mg or 240 mg, or matched placebo, administered every 4 weeks. **Route of Administration:** Subcutaneous injection. **Duration of Treatment:** Up to 3 years (156 weeks) with a 16-week follow-up period.

5.2 Concomitant & Prohibited Therapy Permitted: Standard allergy emergency treatments (e.g., epinephrine auto-injectors, antihistamines) for accidental exposures. **Prohibited:** Concurrent experimental therapies, other monoclonal antibodies, or specific immunosuppressive agents that could confound the assessment of IgE blockade.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
1	Screening	Verification of core study completion, Informed Consent, Rollover
2	Baseline	Day 0, Continuation of treatment allocation, Baseline HRQoL questionnaires
3	Interim	Every 4 Weeks, Subcutaneous study drug administration, Safety assessments, TEAE monitoring
4	End of Study	Up to Year 3 + 16 Weeks, Open-label Oral Food Challenge (OFC), Final Safety Follow-up, HRQoL evaluation

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Long-term safety profile and efficacy assessed by evaluating Treatment-Emergent Adverse Events (TEAEs) and clinical responsiveness. **Measurement Method:**

Incidence of TEAEs (defined as events starting after the first dose of extension treatment up to 16 weeks post-last dose) and Open-label Oral Food Challenge (OFC) / Skin Prick Tests.

7.2 Secondary & Exploratory Endpoints 1. Health-Related Quality of Life (HRQoL) using FAQLQ-TF (evaluating emotional impact, allergen avoidance, dietary restrictions, and risk of accidental exposure). 2. Patient's perceived food allergy severity and risk via the Food Allergy Independent Measure (FAIM-TF/AF).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous evaluation of TEAEs, recorded from the start of the extension study through the 16-week post-treatment follow-up window. **Serious Adverse Events (SAEs):** Specific monitoring for anaphylactic events, severe allergic reactions requiring ICU admission, or severe asthma exacerbations. **Data Monitoring Committee (DMC):** Supervised by the sponsor's internal and independent safety review boards.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set for all participants receiving at least one dose of the extension study drug. **Sample Size Justification:** The sample size ($N=163$) was determined by the number of eligible and consenting participants rolling over from the core ligelizumab Phase III trials. **Handling of Missing Data:** Standard multiple imputation and Last Observation Carried Forward (LOCF) for questionnaire scaling where applicable.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring of clinical sites. **Audit Trail:** Standardized electronic Case Report Forms (eCRF) utilizing verified scoring metrics for patient-reported outcomes (FAQLQ-TF and FAIM-TF).

11. Study Identifiers & Governance

Primary ID: CQGE031G12303B **Registry Number:** NCT05678959 **Sponsor/Lead Organization:** Novartis **Study Title:** Long-term Extension Study of Ligelizumab in Food Allergy **Official Regulatory Title:** A Three-year, Multi-center, Double-blind, Extension Study to Evaluate the Long-term Safety and Efficacy of Ligelizumab in Patients Who Completed Ligelizumab's Phase III Studies in Food Allergy

12. Clinical Framework (Food Allergy Specific)

Primary Condition: Food Allergy **Diagnosis Threshold:** Confirmed IgE-mediated food allergy meeting core Phase III study completion requirements **Medical Methodology:** Anti-IgE Monoclonal Antibody Therapy (Ligelizumab) **Optimization Target:** Increasing the allergen reaction threshold to prevent anaphylaxis from accidental exposure

13. Study Procedures & Methodology

Management Pathway: Subcutaneous injections every 4 weeks **Education Protocol (HCP):** Training on protocol for conditional discontinuation and Open-label Oral Food Challenge (OFC) monitoring **Education Protocol (Patient):** Self-administration training (at home for a specific subset of participants) and strict allergen avoidance adherence **Quality Audit Mechanism:** Standardized scoring via FAQLQ-TF and FAIM-TF instruments to track HRQoL impairment

14. Observation & Follow-Up Schedule

Total Duration: Up to 156 weeks (3 years) plus 16 weeks follow-up **Onsite Visit Cadence:** Every 4 weeks (Q4W) **Remote Monitoring Frequency:** As required for TEAE tracking and self-administration monitoring **Checklist Review Interval:** Regular protocol-defined intervals for HRQoL questionnaires

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				